REPORT: Advanced RAG System

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Title

Building a Production-Level Advanced RAG System using Hybrid Retrieval, Query Expansion, and Re-Ranking

Objective

The objective of this assignment is to build an advanced Retrieval-Augmented Generation (RAG) system that improves over naïve retrieval methods by combining multiple techniques such as:

- Query Expansion
- Hybrid Retrieval (BM25 + SBERT)
- Re-Ranking using Cross-Encoder
- Answer Generation using Gemini

Tools Used

- Python
- HuggingFace (Sentence Transformers)
- Google Gemini API
- rank-bm25
- NumPy

Step-by-Step Implementation

Cell 1: Install Libraries

What we did:
Installed required libraries for embeddings, retrieval, and generation.

Result:
All dependencies installed successfully.

Cell 2: Import Libraries

What we did:
Imported necessary modules like NumPy, BM25, SBERT, CrossEncoder, and Gemini.

Result:
All libraries loaded without errors.

Cell 3: Setup Gemini API

What we did:
Configured API key and loaded Gemini model (gemini-2.5-flash).

Result:
Gemini model initialized successfully and ready for use.

Cell 4: Create Document Corpus

What we did:
Created a dataset of 20 AI/ML-related documents.

Key Features:

- Multiple documents per topic
- Included technical terms (BERT, AdamW, etc.)

Result:
Corpus successfully created and ready for retrieval.

Cell 5: Load Models

What we did:
Loaded:

- SBERT model → for semantic embeddings
- Cross-Encoder → for re-ranking

Result:
Models loaded successfully.

Cell 6: Test SBERT

What we did:
Generated embedding for a query.

Result:

```
Embedding shape: (384,)
```

Explanation:
Each query is converted into a 384-dimensional vector for semantic comparison.

Cell 7: Setup BM25

What we did:
Tokenized corpus and initialized BM25.

Result:
BM25 ready for keyword-based retrieval.

Cell 8: Test BM25

What we did:
Tested BM25 using a sample query.

Result:
Correct document retrieved based on keywords.

Cell 9: Hybrid Retriever

What we did:
Implemented hybrid retrieval combining:

- BM25 (keyword search)
- SBERT (semantic search)

Used **RRF (Reciprocal Rank Fusion)** to merge results.

Result:
Returned documents with:

- BM25 rank
- SBERT rank
- RRF score

Cell 10: Test Hybrid Retrieval

What we did:
Tested hybrid retrieval.

Sample Output:

bm25_rank: 1, sbert_rank: 1 → best match

Result:
Hybrid retrieval successfully combined both methods.

Cell 11: Re-Ranker

What we did:
Implemented cross-encoder re-ranking.

Result:
Assigned scores to documents based on relevance.

Cell 12: Test Re-Ranker

What we did:
Tested re-ranking.

Sample Output:

-0.026 → best match
-11 → less relevant

Explanation:
Higher score (less negative) means more relevant.

Cell 13: Query Expansion

What we did:
Used Gemini to generate 3 variations of query.

Example:

what is attention
explain attention mechanism
attention in machine learning

Result:
Improved ability to handle vague queries.

Cell 14: Test Query Expansion

What we did:
Verified generated queries.

Result:
Multiple meaningful query variations generated.

Cell 15: Advanced RAG Pipeline

What we did:
Built full pipeline:

1. Query Expansion
2. Hybrid Retrieval
3. Re-Ranking
4. Gemini Answer Generation

Result:
Complete end-to-end system created.

Cell 16: Test Advanced RAG

What we did:
Tested full pipeline.

Result:
System generated:

- Final answer
- Top relevant documents

Cell 17: Naïve RAG

What we did:
Implemented baseline using only SBERT.

Result:
Simple retrieval without improvements.

Cell 18: Comparison

What we did:
Compared Naïve vs Advanced RAG.

Cell 19: Results Table

Final Output:

Query	Naive RAG Top Doc	Advanced RAG Top Doc	Different?
optimization techniques for training	Batch normalization stabilizes training	Gradient descent minimizes loss	Yes
what is overfitting	Overfitting occurs when model memorizes data	Same	No
transformers encode meaning	Positional encoding helps	Same	No

Observations

- Advanced RAG performs better for complex queries
- Query expansion improves understanding of vague queries
- Hybrid retrieval combines strengths of keyword and semantic search

- Re-ranking improves relevance of final results

Conclusion

The Advanced RAG system significantly improves retrieval quality compared to Naïve RAG by incorporating:

- Query expansion
- Hybrid retrieval
- Re-ranking

This approach is more robust and effective for real-world applications where queries are often vague.